

```
1 grammar = { 'S': [['NP', 'VP']], 'NP': [['Det', 'N']], 'VP': [['V', 'NP']], 'Det': [['the'], ['a']], 'N': [['cat'], ['dog']], 'V': [['sees'], ['likes']] }
2 sentence = input("Enter sentence: ").lower().split()
3 def parse(symbol, words, pos):
4     if symbol not in grammar:
5         if pos < len(words) and symbol == words[pos]:
6             return pos + 1
7         return None
8     for rule in grammar[symbol]:
9         current = pos
10        success = True
11        for item in rule:
12            result = parse(item, words, current)
13            if result is None:
14                success = False
15                break
16            current = result
17        if success:
18            return current
19    return None
20 result = parse(symbol: 'S', sentence, pos: 0)
21 if result == len(sentence):
22     print("Sentence is accepted")
23 else:
24     print("Sentence is rejected")
```

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/Users/moshika/PyCharmMiscProject/.venv/bin/python /Users/moshika/PyCharmMiscProject/Exp 11.py
Enter sentence: the cat sees a dog
Sentence is accepted
```

